

120 DSA Interview Questions (Pattern Based Guide)

1. Arrays & Basic Logic (15 Questions)

- Two Sum
- Best Time to Buy and Sell Stock
- Maximum Subarray (Kadane's Algorithm)
- Contains Duplicate
- Product of Array Except Self
- Rotate Array
- Merge Sorted Arrays
- Move Zeroes
- Find Missing Number
- Majority Element
- Intersection of Two Arrays
- Sort Colors (Dutch National Flag)
- Longest Consecutive Sequence
- Subarray Sum Equals K
- Find Pivot Index

2. Strings (15 Questions)

- Valid Anagram
- Valid Palindrome
- Longest Substring Without Repeating Characters
- Group Anagrams
- Longest Palindromic Substring
- String to Integer (atoi)
- Implement strStr()
- Minimum Window Substring
- Encode and Decode Strings
- Count and Say
- Reverse Words in a String
- Check if Strings are Rotations
- Roman to Integer
- Zigzag Conversion
- Multiply Strings

3. Two Pointers & Sliding Window (15 Questions)

- Container With Most Water
- 3Sum
- Remove Duplicates from Sorted Array
- Trapping Rain Water
- Minimum Size Subarray Sum
- Longest Repeating Character Replacement
- Permutation in String
- Find All Anagrams in a String
- Sliding Window Maximum
- Max Consecutive Ones
- Subarrays with K Different Integers
- Fruit Into Baskets
- Smallest Subarray with Given Sum
- Binary Subarrays With Sum
- Longest Subarray of 1's After Deleting One Element

4. Binary Search (10 Questions)

- Binary Search
- Search in Rotated Sorted Array
- Find Minimum in Rotated Sorted Array
- Search Insert Position
- First Bad Version
- Koko Eating Bananas
- Capacity to Ship Packages Within D Days
- Find Peak Element
- Median of Two Sorted Arrays
- Count Negative Numbers in Sorted Matrix

5. Stack & Queue (10 Questions)

- Valid Parentheses
- Min Stack
- Evaluate Reverse Polish Notation
- Daily Temperatures
- Next Greater Element
- Implement Queue using Stacks
- Implement Stack using Queues
- Largest Rectangle in Histogram
- Sliding Window Maximum (Deque)
- Asteroid Collision

6. Linked List (10 Questions)

- Reverse Linked List
- Merge Two Sorted Lists
- Linked List Cycle
- Remove Nth Node From End
- Add Two Numbers
- Intersection of Two Linked Lists
- Palindrome Linked List
- Reorder List
- Copy List with Random Pointer
- LRU Cache

7. Trees & BFS/DFS (20 Questions)

- Maximum Depth of Binary Tree
- Same Tree
- Invert Binary Tree
- Binary Tree Level Order Traversal
- Validate Binary Search Tree
- Lowest Common Ancestor of BST
- Diameter of Binary Tree
- Balanced Binary Tree
- Path Sum
- Binary Tree Right Side View
- Construct Binary Tree from Preorder and Inorder
- Serialize and Deserialize Binary Tree
- Kth Smallest Element in BST
- Subtree of Another Tree
- Count Good Nodes in Binary Tree
- Number of Islands
- Clone Graph
- Course Schedule
- Word Ladder
- Rotting Oranges

8. Greedy & Prefix Sum (10 Questions)

- Jump Game
- Jump Game II
- Gas Station
- Candy
- Task Scheduler
- Merge Intervals
- Insert Interval
- Meeting Rooms
- Non-overlapping Intervals
- Maximum Product Subarray

9. Dynamic Programming (15 Questions)

- Climbing Stairs
- House Robber
- House Robber II
- Coin Change
- Longest Increasing Subsequence
- Longest Common Subsequence
- 0/1 Knapsack
- Partition Equal Subset Sum
- Unique Paths
- Minimum Path Sum
- Decode Ways
- Word Break
- Edit Distance
- Maximum Profit in Job Scheduling
- Target Sum